

Chapter Six

Output

Discovering Computers 2012

Your Interactive Guide to
the Digital World



Objectives Overview

Describe the types of output

Explain the characteristics of various display devices

Describe the factors that affect the quality of an LCD monitor or LCD screen

Describe various ways to print

Differentiate between a nonimpact printer and an impact printer

Objectives Overview

Summarize the characteristics of ink-jet printers, photo printers, laser printers, multifunction peripherals, thermal printers, mobile printers, label and postage printers, and plotters and large-format printers

Describe the uses and characteristics of speakers, headphones, and earbuds

Identify the purpose and features of data projectors, interactive whiteboards, and force-feedback game controllers and tactile output

Identify output options for physically challenged users

What Is Output?

- **Output** is data that has been processed into a useful form



What Is Output?

- An **output device** is any type of hardware component that conveys information to one or more people

Display devices

Printers

Speakers,
headphones,
and earbuds

Data projectors

Interactive
whiteboards

Force-feedback
game
controllers

Tactile output

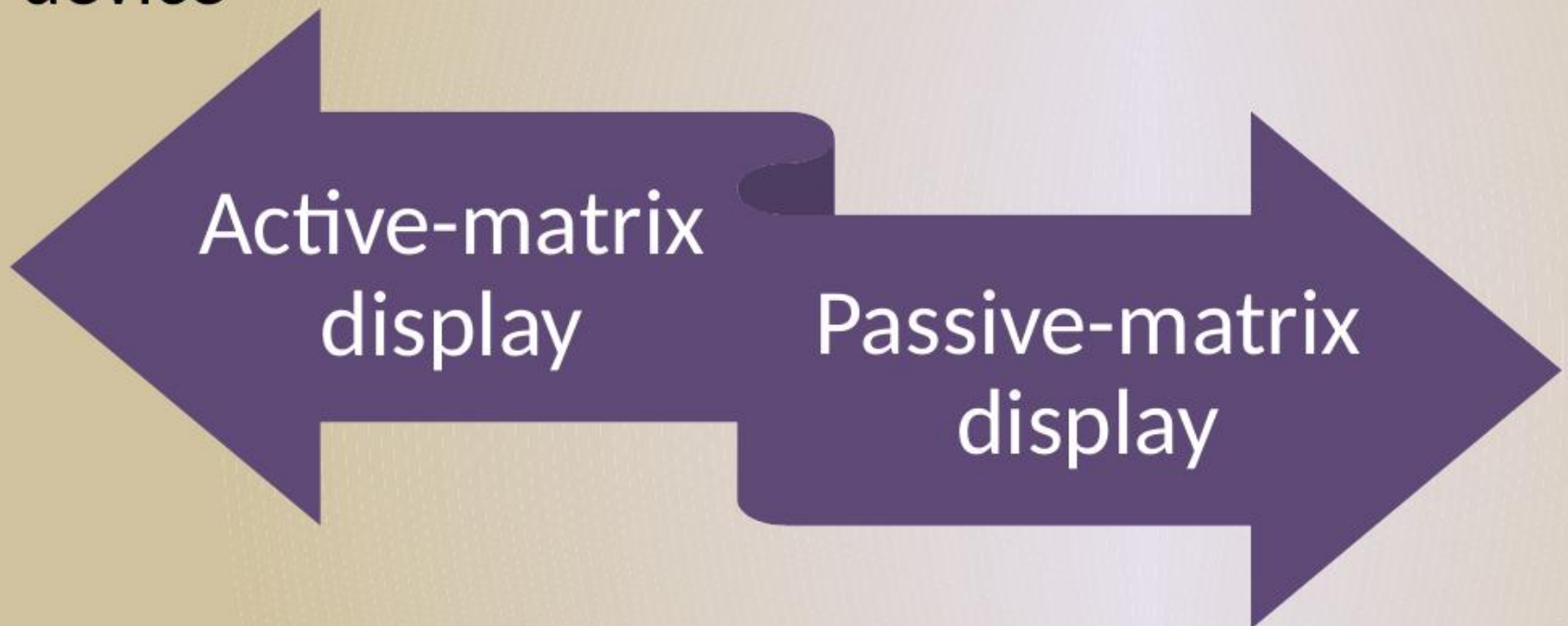
Display Devices

- A **display device** visually conveys text, graphics, and video information
- A **monitor** is packaged as a separate peripheral
 - **LCD monitor**
 - Widescreen



Display Devices

- **Liquid crystal display (LCD)** uses a liquid compound to present information on a display device



Display Devices

- The quality of an LCD monitor or LCD screen depends primarily on:



Resolution

Response time

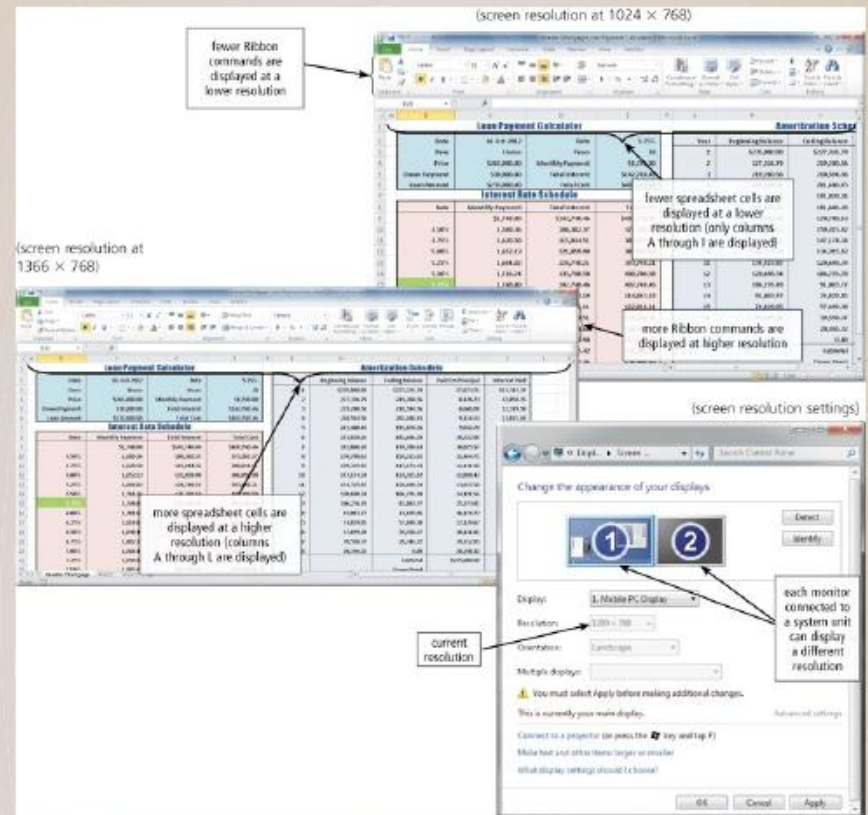
Brightness

Dot pitch

Contrast ratio

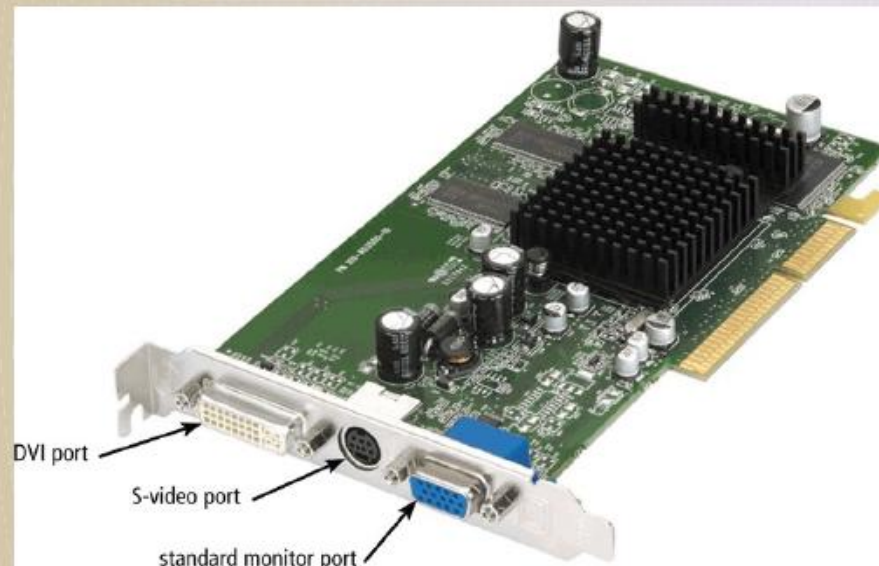
Display Devices

- **Resolution** is the number of horizontal and vertical pixels in a display device
 - A higher resolution uses a greater number of pixels



Display Devices

- The graphics processing unit (GPU) controls the manipulation and display of graphics on a display device
- LCD monitors use a digital signal and should plug into a DVI port, an HDMI port, or a DisplayPort



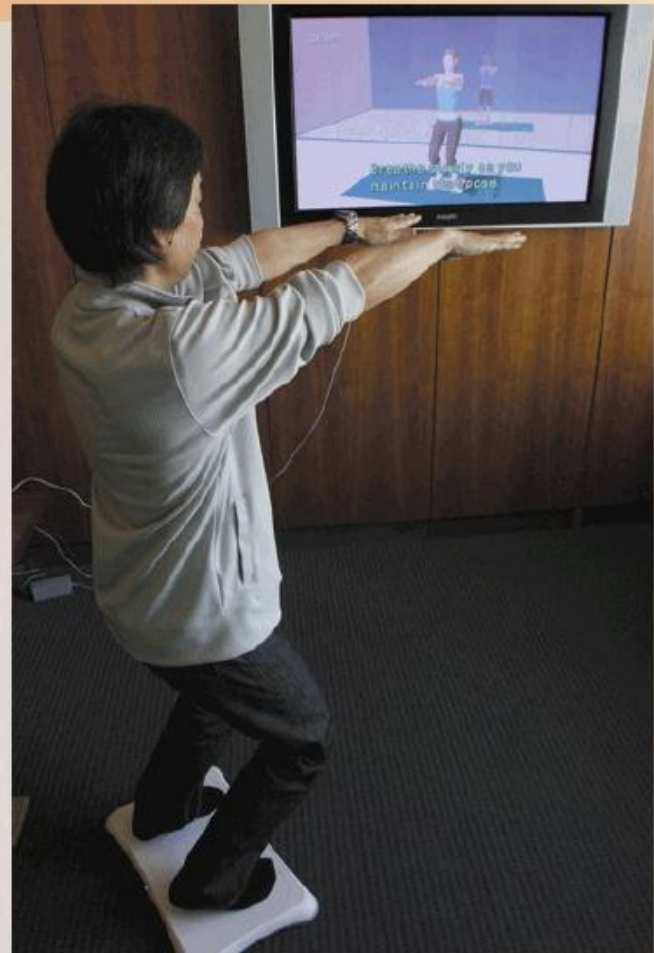
Display Devices

- **Plasma monitors** are display devices that use gas plasma technology and offer screen sizes up to 150 inches



Display Devices

- Televisions also are a good output device
 - Require a converter if you are connecting your computer to an analog television
- Digital television (DTV) offers a crisper, higher-quality output
- **HDTV** is the most advanced form of digital television



Display Devices

- A **CRT monitor** is a desktop monitor that contains a cathode-ray tube
 - Have a much larger footprint than do LCD monitors



Printers

- A **printer** produces text and graphics on a physical medium
 - Printed information is called a hard copy, or printout
 - Landscape or portrait orientation

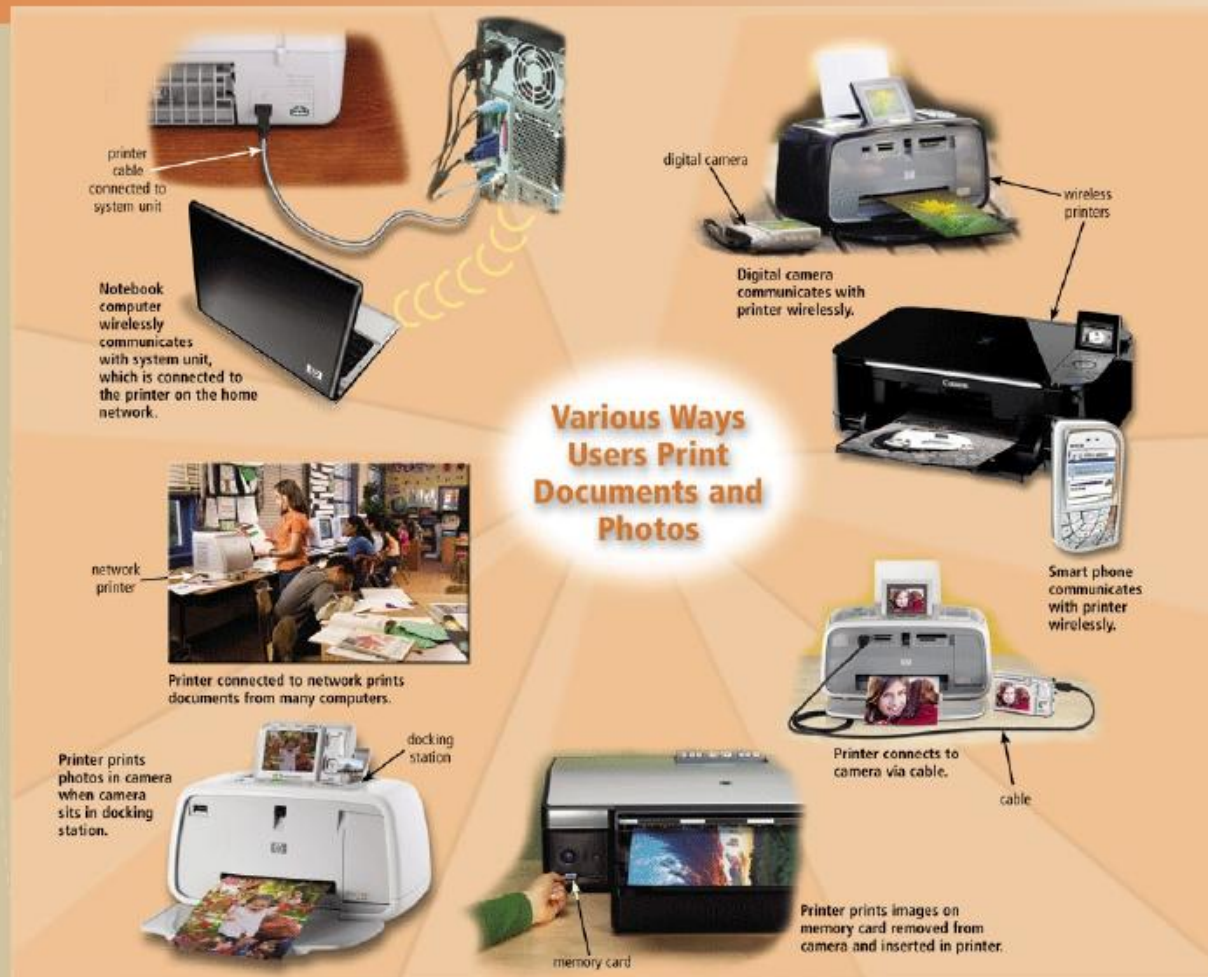


Printers

Before purchasing a printer, ask yourself a series of questions:

1. *What is my budget?*
2. *How fast must my printer print?*
3. *Do I need a color printer?*
4. *What is the cost per page for printing?*
5. *Do I need multiple copies of documents?*
6. *Will I print graphics?*
7. *Do I want to print photos?*
8. *Do I want to print directly from a memory card?*
9. *What types of paper does the printer use?*
10. *What sizes of paper does the printer accept?*
11. *Do I want to print on both sides of the paper?*
12. *How much paper can the printer tray hold?*
13. *Will the printer work with my computer and software?*
14. *How much do supplies such as ink, toner, and paper cost?*
15. *Can the printer print on envelopes?*
16. *How many envelopes can the printer print at a time?*
17. *How much do I print now, and how much will I be printing in a year or two?*
18. *Will the printer be connected to a network?*
19. *Do I want wireless printing capability?*

Printers



Printers

- A **nonimpact printer** forms characters and graphics on a piece of paper without actually striking the paper

Ink-jet
printers

Photo
printers

Laser
printers

Thermal
printers

Mobile
printers

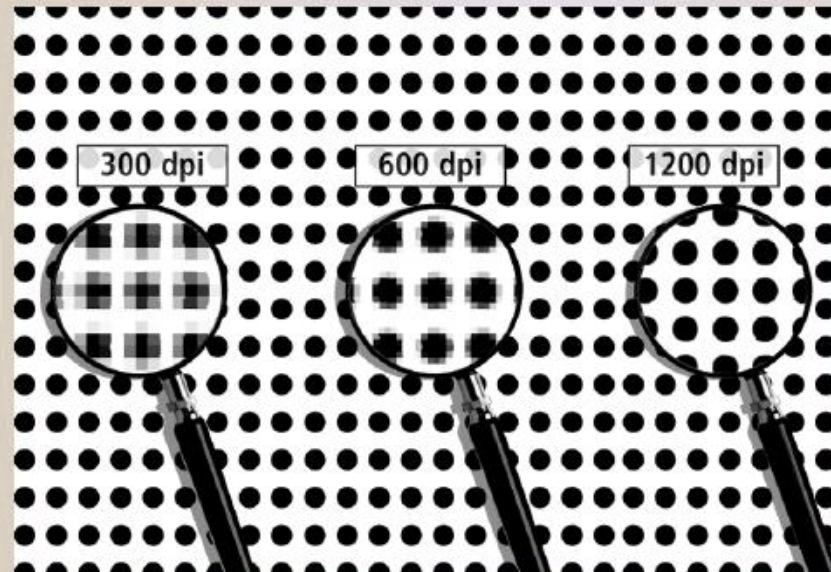
Label and
postage
printers

Plotters

Large-
format
printers

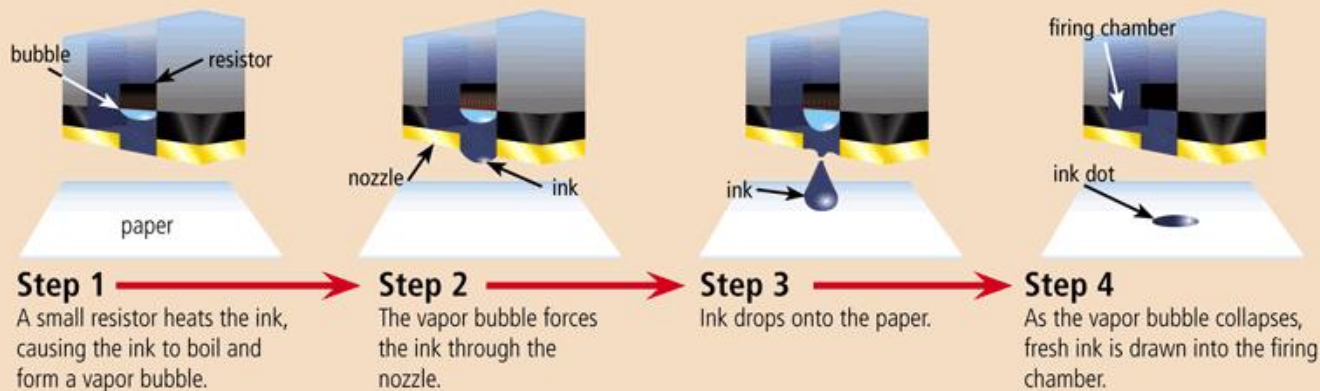
Printers

- An **ink-jet printer** forms characters and graphics by spraying tiny drops of liquid ink onto a piece of paper
 - Color or black-and-white
 - Printers with a higher dpi (dots per inch) produce a higher quality output



Printers

How an Ink-Jet Printer Works



Printers



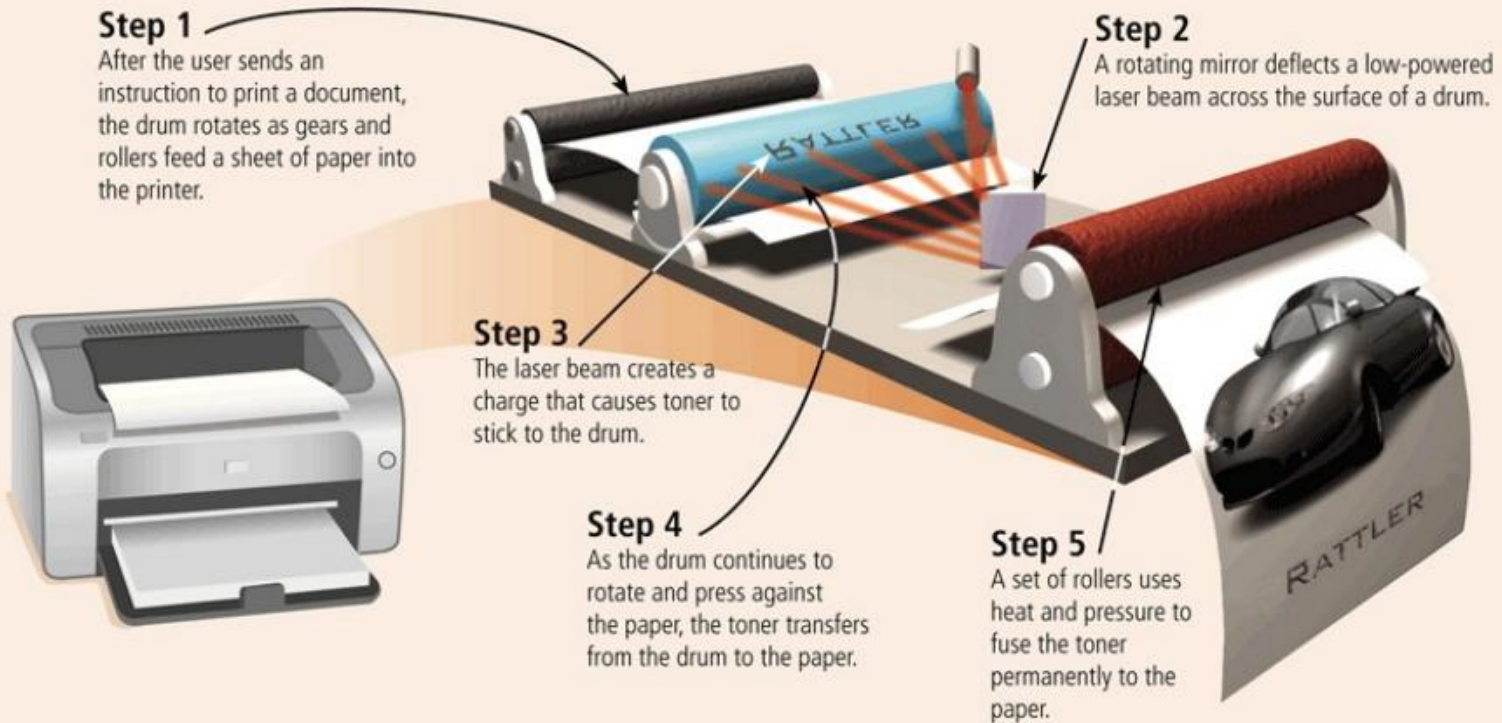
Printers



Printers

Printers

How a Black-and-White Laser Printer Works



Printers

- A **multifunction peripheral** (MFP) is a single device that prints, scans, copies, and in some cases, faxes
 - Sometimes called an all-in-one device



Printers

- A **thermal printer** generates images by pushing electrically heated pins against the heat-sensitive paper



Printers

- A **mobile printer** is a small, lightweight, battery-powered printer that allows a mobile user to print from a notebook computer, smart phone, or other mobile device



Printers

- A **label printer** is a small printer that prints on adhesive-type material
- A postage printer prints postage stamps
 - Postage also can be printed on other types of printers



Printers

- **Plotters** are used to produce high-quality drawings
- **Large-format printers** create photo-realistic quality color prints on a larger scale

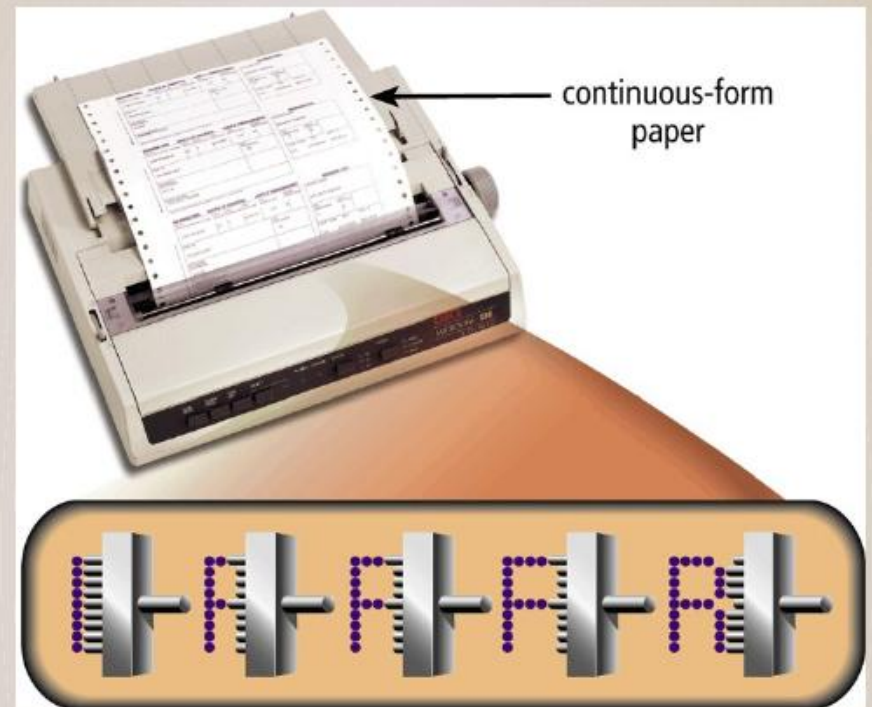


Printers

- **Impact printers** form characters and graphics on a piece of paper by striking a mechanism against an inked ribbon that physically contacts the paper

Printers

- A **dot-matrix printer** produces printed images when tiny wire pins on a print head mechanism strike an inked ribbon
- A **line printer** prints an entire line at a time



Speakers, Headphones, and Earbuds

- An **audio output device** produces music, speech, or other sounds

Speakers, Headphones, and Earbuds

- **Headphones** are speakers that cover or are placed outside of the ear
- **Earbuds** (also called earphones) rest inside the ear canal



Speakers, Headphones, and Earbuds

- Some speakers are specifically designed to play audio from a portable media player
 - Wireless speakers



Speakers, Headphones, and Earbuds

Other Output Devices

- Other output devices are available for specific uses and applications

Other Output Devices

- A **data projector** is a device that takes the text and images displaying on a computer screen and projects them on a larger screen
 - LCD projector
 - Digital light processing (DLP) projector



Other Output Devices

- An **interactive whiteboard** is a touch-sensitive device, resembling a dry-erase board, that displays the image on a connected computer screen



Other Output Devices

- Force-feedback sends resistance to the device in response to actions of the user
- Tactile output provides the user with a physical response from the device



Putting It All Together

Suggested Output Devices by User

User

Monitor

Printer

Other

Home



- 19- or 20-inch LCD monitor, or 17-inch LCD screen on notebook computer

- Ink-jet color printer; or
- Photo printer

- Speakers
- Headphones or earbuds or headset
- Force-feedback game controller or tactile output

Small Office/Home Office



- 20- or 22-inch LCD monitor
- LCD screen smart phone or other mobile device

- Multifunction peripheral; or
- Ink-jet color printer; or
- Laser printer (black-and-white or color)
- Label printer
- Postage printer

- Speakers

Putting It All Together

Suggested Output Devices by User

User

Mobile



Monitor

- 17-inch LCD screen on notebook computer
- 8.9-inch screen on a netbook
- LCD screen on smart phone or other mobile device

Printer

- Mobile color printer
- Ink-jet color printer; or
- Laser printer for in-office use (black-and-white or color)
- Photo printer

Other

- Headphones or earbuds or headset
- DLP data projector

Power



- 30-inch LCD monitor

- Laser printer (black-and-white or color)
- Plotter or large-format printer; or
- Photo printer; or
- Dye-sublimation printer

- Speakers
- Headphones or earbuds or headset

Putting It All Together

Suggested Output Devices by User

User

Monitor

Printer

Other

Enterprise

- 20- or 22-inch LCD monitor
- LCD screen on smart phone or other mobile device

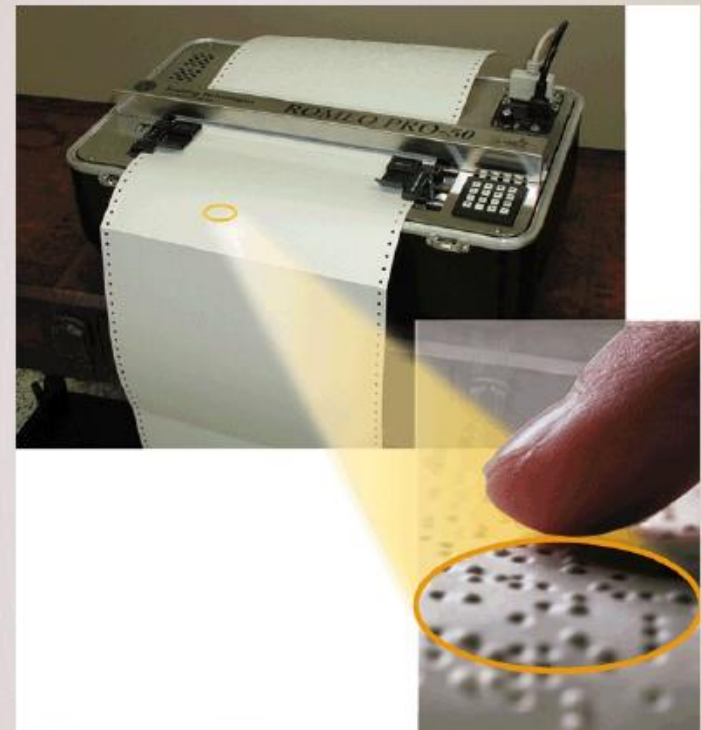
- High-speed laser printer
- Laser printer, color
- Line printer (for large reports from a mainframe)
- Label printer

- Speakers
- Headphones or earbuds or headset
- Networked DLP data projector
- Interactive whiteboard



Output Devices for Physically Challenged Users

- Many accessibility options exist for users with various disabilities



Summary

Chapter Six

Output

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the Digital World

Chapter 6 Complete

